



KARTIK RAJ

Final Year Undergraduate
Department of Mechanical Engineering
Indian Institute of Technology Kanpur

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Examination	Degree/Certificate	Institute	Year	CPI/%
Graduation	B.Tech	Indian Institute of Technology Kanpur	2023-2027	8.0/10
Intermediate/+2	CBSE (XII)	S.K.P. Vidya Vihar, Deoghar	2023	95.2%
Matriculation	CBSE (X)	D.A.V. Public School, Deoghar	2021	95.2%

SCHOLASTIC ACHIEVEMENTS

- **National 1st Runner-up** in **American Express CodeStreet '26**, selected from **15,000+** competing registrations nationwide (2026)
- Recipient of **Pritam Lal Shakuntala Rawal Memorial Scholarship** for good academic performance consistently (2024, 2025, 2026)
- Winner of **AI CityLens Hackathon**, organized by **Airawat Foundation** among **250+** participants nationwide (2026)
- Ranked **Top 100** nationally in **InMobi Codies Challenge'26** (2026)

WORK EXPERIENCE

I'mBesideYou Inc., Japan

May '26 – Jul '26

Software Development Engineering Intern

- Engineered **AI-powered** automation to autonomously build and maintain a live multi-channel sales pipeline for a **119-company** portfolio, eliminating manual CRM upkeep via **LLM** inference
- Widely used **AWS ECS**, **FastAPI**, **Firestore** (security rules, composite indexes) & **OAuth 2.0** alongside React & Typescript
- Built fault-tolerant **APScheduler** workers, reconciliation jobs, **asyncio.Lock** based single-instance locking & outage self-recovery
- Containerized the service with **Docker** and cut third-party API cost **85%** through incremental batching & dynamic rate-capping
- Designed **event-driven ingestion** across Gmail, Zoom, Plaud & Messenger using **Pub/Sub** webhooks with **idempotent replay**

Byufuel India Pvt. Ltd.

Jan '26 – Apr '26

Founder's Office Intern

- Modeled multi-objective routing minimizing **cost** and **carbon emissions** by **18-25%**, via **greedy heuristic** for **NP-hard CVRPTW**
- Used a **priority queue** to greedily select stops by cost, weighted on travel time, waiting penalty and enabled mid-route rerouting
- Cached distance-matrix lookups from **OpenRouteService** with a **Haversine** fallback to always avoid repeat shortest-path queries
- Built a **normalized relational schema** (3NF) for fleets, warehouses & restaurants in **PostgreSQL** at **2,000+ MT** scale across **50+ cities**

TerrAqua UAV Pvt. Ltd.

Dec '25

WebGIS Developer Intern

- Engineered a **WebGIS** interface (**React**, **Leaflet**) with real-time tracking, persisting drone paths and waypoints in **PostgreSQL**
- Built an offline-first **Android** app via **Capacitor**, reliably syncing field data to the backend on reconnect in low-connectivity zones
- Deployed a **Node.js** backend with **Cloudinary** for geo-tagged image storage, adding **GeoJSON/KML** export for interoperability

TECHNICAL SKILLS

- **Languages:** Python, SQL, C/C++, JavaScript (ES6+), HTML, CSS
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, TensorFlow, PyTorch, LangChain, LangGraph, ChromaDB
- **Frameworks/Tools:** React.js, Node.js, Express.js, Redis, Postman, Vercel, FastAPI, PostgreSQL, Firebase, Git/GitHub, Docker, AWS

RELEVANT COURSEWORK

(* Online)

Intro to Computer Networking*	Intro to Operating Systems*
Database Management Systems*	System Design Basics*
Introduction to Machine Learning	Intro to Linear Algebra
Computer Vision & Deep Learning	Ordinary Differential Eqns.

POSITIONS OF RESPONSIBILITY

- **Head | ShARE IITK:** Spearheaded a **30+** associate team across a **4-tier** structure, managing an **INR 1 Lakh+** budget

KEY COMPETITIONS

American Express CodeStreet'26 | 1st Runner-up Jul–Aug'26

AEGIS – Governance Layer for Autonomous AI Payment Agents

- Governed autonomous AI payment agents via a **Redis** token-bucket limiter, using atomic **INCR** preventing race conditions
- Decided every agent action through a **Cedar-style** deterministic policy evaluation engine under a **p99 sub-50ms** decision budget
- Blocked privilege escalation at credential issuance via **subset-scoped delegation** on signed **JWS/SD-JWT** mandates
- Proved tamper-evidence with a **SHA-256 hash-chained** ledger, localizing any break to exact broken link via **Merkle** checkpoints

Inter IIT Tech Meet 14.0 | 4th of 23 IITs

Oct'25 – Dec'25

Pathway - Operational Intelligence Platform | IIT Patna

- Compiled drag-and-drop pipelines into a **Pathway** streaming **DAG**, guaranteeing **exactly-once** processing via topological sort
- Orchestrated per-pipeline **Docker** containers with dynamic database credentials, isolating each pipeline on its own network
- Secured agent tool calls via a **zero-trust**, **HMAC**-signed RPC layer enforcing **SSRF** checks, rate limiting and role-based access control
- Implemented workflow **checkpointing** with auto-saved pipeline versions, to let any run roll back instantly to a prior state on failure
- Built an **extensible** schema-driven plugin system mapping custom node types to **Pathway** transform functions, extending **40+** built-in connectors without core engine changes

AI CityLens Hackathon, Airawat Foundation | Winner

Jun '26

CityLens – Urban Hazard Detection Pipeline

- Built an end-to-end custom **YOLOv8** pipeline to detect **5 hazard classes** from street imagery, winning the hackathon outright
- Curated a multi-source hazard dataset, standardizing label formats and class definitions across heterogeneous sources
- Achieved **0.97 mAP@0.5** on high-signal classes like fire/smoke, identifying low-light hazard detection as the key limiting factor

PROJECTS

Saliency Detection in Remote Sensing Images Mar '26 – Apr '26

Course Project | EE655 | Mentor: Prof. Koteswar Rao

- Benchmarked **5 deep learning architectures** for salient object detection on aerial imagery evenly across **8 evaluation metrics**
- Found **BASNet** optimal for boundary-critical tasks (highest **IoU 0.82**) versus **RSSODNet** for recall-critical satellite tasks (**0.8957**)
- Traded off **DeepLabV3+**'s lowest pixel error (**MAE 0.011**) against measurably weaker recall to inform task-dependent model choice
- Trained all 5 models independently in **PyTorch** under one evaluation protocol to isolate architecture effects from variance

Bayesian Inference & Model Comparison

May '26 – June '26

Course Project | CGS698C | Prof. Himanshu Yadav

- Implemented **Bayesian linear** and **Poisson regression** models with prior sensitivity analysis and posterior predictive checks
- Built **Gibbs** & **Metropolis-Hastings** samplers alongside **Hami-tonian Monte Carlo**, diagnosing convergence via trace plots
- Compared models via **LOO-CV** across **2** priors, to cut deviance from **54.5** to **42.2** ultimately and reversing the in-sample ranking

ML-PUF Modeling & PUF Attacks

Jun '26 – Jul '26

Course Project | CS771 | Prof. Purushottam Kar

- Engineered a **105-dimensional** feature map exploiting underlying XOR-mix structure, making **PUF** response pairs linearly separable
- Recovered **256** intrinsic arbiter delays from **65** observed parameters by solving an underdetermined linear reconstruction
- Generalized the attack to **Hamming** and **Bit-in-the-Middle PUFs** via sign-flip feature tricks, each exceeding **99%** test accuracy